

# STANDARD OPERATING PROCEDURE — BEARING REPLACEMENT

SOP Number: SOP-MECH-007 | Revision: Rev 4 | Applies to: All rotating equipment

Compliance: OISD-113, Factory Act Section 7A, ISO 10816-3

## 1. Prerequisites and Safety

- 1.1 Obtain Permit to Work (PTW) from area supervisor before beginning work.
- 1.2 Apply LOTO (Lock-Out Tag-Out) on motor M-104 at MCC panel. Verify zero energy state.
- 1.3 Gas test required if work is in a classified hazardous area. Confirm with safety officer.
- 1.4 PPE: Safety boots, gloves, eye protection, hard hat. Hearing protection during operation checks.

## 2. Bearing Replacement Procedure

- Step 1: Drain coupling housing. Remove coupling guard and coupling half from pump shaft.
- Step 2: Remove bearing housing end covers. Extract bearing using puller tool. Do not use hammer on bearing — shaft damage risk.
- Step 3: Clean shaft journal with lint-free cloth. Inspect for scoring or step wear. Maximum allowable shaft taper: 0.02 mm. If exceeded, replace shaft.
- Step 4: Heat new bearing to 80°C in oil bath or induction heater. Never exceed 120°C. Install immediately while warm. Do not use cold press-fit.
- Step 5: Apply 15g of Mobil Grease XHP 222 to each bearing housing. Do not over-lubricate — excess grease causes overheating.
- Step 6: Reassemble in reverse order. Check axial float: 0.1 – 0.3 mm acceptable.

## 3. Post-Replacement Verification

- 3.1 Check coupling alignment: angular < 0.05 mm/100mm, parallel < 0.05 mm.
- 3.2 Run pump unloaded for 30 minutes. Monitor vibration and temperature continuously.
- 3.3 Vibration must be < 2.3 mm/s before returning to full load service.
- 3.4 Record work order completion in CMMS. Set lubrication reminder at +500 hours.